EXPERIMENT-10

Aim: Implementing Notification. Build an android application that sends notification to the user. Customize the appearance and behaviour of the notifications.

Theory:
Implementing notifications in Android involves using the **Notification Manager** and **Notification Channels** to deliver alerts to users even when the application is not actively in use. A notification is built using the **NotificationCompat.Builder** class, where developers can define elements such as title, message, icon, priority, and actions. For devices running Android 8.0 (API level 26) and above, **notification channels** are required to control the importance and behavior of notifications, such as sound, vibration, and visibility. Developers can customize the appearance using styles like BigTextStyle or InboxStyle, and enhance interaction by adding action buttons or pending intents to open specific activities. Proper handling of runtime permissions (like POST_NOTIFICATIONS in newer Android versions) ensures that notifications are displayed effectively while maintaining user privacy and control.

App Notifications Disabled (Device Setting) - Sometimes phone blocks notifications.
✓ Fix:
1. Go to Settings
2. Apps → Your App
3. Notifications → Turn ON

Use of POST_NOTIFICATIONS permission: This permission allows your app to "Send notifications to the user"
Why is it important?
Starting from Android 13 (API 33), Apps cannot show notifications without user permission. User must allow notification manually.
What happens without this permission?
Notification code runs but nothing is displayed.

Step 1: Create New Project
1. Open Android Studio
2. Click New Project
3. Select Empty Views Activity
4. Name: NotificationApp
5. Language: Java (or Kotlin if you prefer)
6. Click Finish

Step 2: Open res/layout/activity_main.xml and write code for widgets.

Step 3: Add Notification Code in MainActivity.

Step 4: Customize Notification
You can customize like this:

.setContentTitle("New Message ")
.setContentText("You have a new notification")
.setSmallIcon(android.R.drawable.star_on)
.setAutoCancel(true)
Add Sound / Vibration:
.setDefaults(NotificationCompat.DEFAULT_ALL)

Step 5: Add this in **AndroidManifest.xml**
<uses-permission android:name="android.permission.POST_NOTIFICATIONS" />

Note: - If nothing is displayed when you click the “Send Notification” button, don’t worry—this is a very common issue.

Code:
activity_main.xml

activity_main.xml × MainActivity.java AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btnNotify"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Notification"
        android:layout_centerInParent="true"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

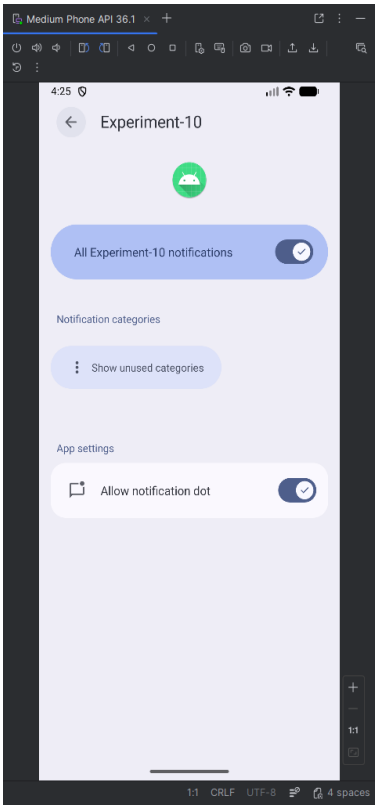
activity_main.xml | MainActivity.java | AndroidManifest.xml

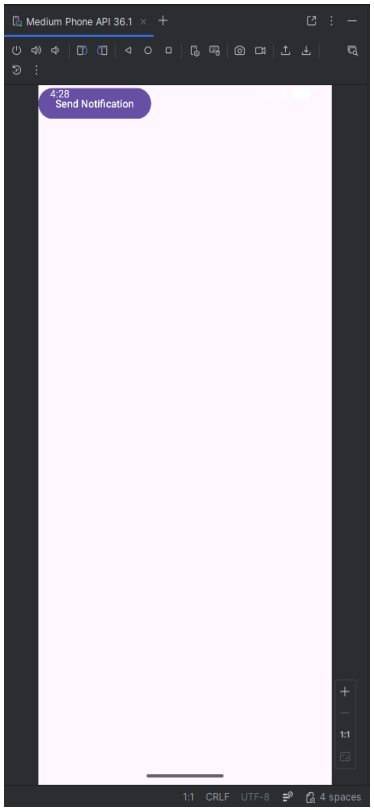
```java
package com.example.experiment_10;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.os.Build;
import android.view.View;
import android.widget.Button;
import androidx.core.app.NotificationCompat;

public class MainActivity extends AppCompatActivity {
    2 usages
    Button btnNotify;
    2 usages
    String CHANNEL_ID = "my_channel_id";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnNotify = findViewById(R.id.btnNotify);
        createNotificationChannel();
        btnNotify.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                NotificationCompat.Builder builder =
                        new NotificationCompat.Builder( context: MainActivity.this, CHANNEL_ID)
                                .setSmallIcon(android.R.drawable.ic_dialog_info)
                                .setContentTitle("Hello Student ")
                                .setContentText("This is your first notification!")
                                .setPriority(NotificationCompat.PRIORITY_HIGH)
                                .setContentTitle("New Message ")
                                .setContentText("You have a new notification")
                                .setSmallIcon(android.R.drawable.star_on)
                                .setAutoCancel(true)
                                .setDefaults(NotificationCompat.DEFAULT_ALL);
                NotificationManager manager = (NotificationManager)
                        getSystemService(NOTIFICATION_SERVICE);
                manager.notify( id: 1, builder.build());
            }
        });
    }
    1 usage
    private void createNotificationChannel() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            NotificationChannel channel = new NotificationChannel(
                    CHANNEL_ID,
                    name: "My Channel",
                    NotificationManager.IMPORTANCE_HIGH
            );
            NotificationManager manager =
                    getSystemService(NotificationManager.class);
            manager.createNotificationChannel(channel);
        }
    }
}
```

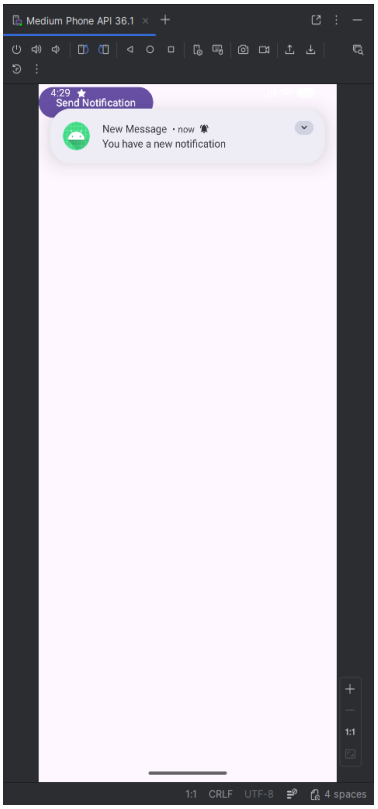
AndroidMenifest.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.POST_NOTIFICATIONS" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="Experiment-10"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Experiment10">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Output:







Conclusion:

This experiment demonstrated how to create and display notifications in an Android application using the Notification Manager. It also highlighted how to customize notification appearance and behavior using channels, styles, and priorities. Overall, it improved understanding of user engagement through effective alert mechanisms in Android apps.